

# OMAR IBRAHIM

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## Education

### Carleton University

Sep. 2024 – Present

*Bachelor of Engineering in Software Engineering*

*Ottawa, Ontario*

- **Relevant Coursework:** Data Structures, Digital Systems, Computer Architecture, Software Engineering

## Technical Skills

**Languages:** Python, C++, TypeScript, SQL, Bash, Java

**Technologies:** Linux, Next.js, PostgreSQL, Supabase, Nginx, Cloudflare, Tailwind CSS, Redis, React

**Developer Tools:** Git, Docker, VS Code, Kali Linux, Burp Suite, WSL, Postman

**AI/ML:** Computer Vision, YOLOv8, OpenCV, Ollama, Agentic AI, LLM Integration, RAG, Prompt Engineering

**Practices:** CI/CD, Agile/Scrum, Secure Coding, OOP, API Design, Unit Testing, UML Modeling, Penetration Testing

## Experience

### Software Developer Student

September 2025 – April 2026

*University of Ottawa MSA*

*Ottawa, Ontario*

- Developed [uomsa.ca](https://uomsa.ca) in **Next.js**, eliminating GoDaddy builder and **cutting annual costs by \$500**.
- Worked on an **8-developer Agile team** to build a library system supporting **1,000+ students in 5 organizations**.
- Built frontend **React components** and pushed migrations to **PostgreSQL** database for system's admin dashboard.

### Software Developer

October 2025 – January 2026

*Competitech Inc.*

*Ottawa, Ontario*

- Worked in an Agile team to build a cloud **AI dialogue system** that plugs into game engines through **custom SDKs**.
- Diagnosed custom backend and **REST APIs** to discover **6 critical flaws** using **Burp Suite**, **Postman**, and **cURL**.
- Fixed a payment issue through **Stripe API** that bypassed payments and granted subscriptions **in under 5 hours**.

### Technical Workshop Lead

March 2026

*Institute of Electrical and Electronics Engineers (IEEE)*

*Ottawa, Ontario*

- Delivered **4 technical workshops** to **100+ students** across campus for a 2-week workshop series at uOttawa.
- Taught **Linux** fundamentals, **Docker**, **Github Actions** CI/CD, **AI/Machine Learning**, and **Cybersecurity**.
- Built a **lab environment** with isolated per-student **Docker containers** and **custom challenges** to test students.

## Projects

🔗 **Hackathon Winner – Revenant** | *Python, Next.js, FastAPI, Redis, PostgreSQL, RAG*

March 2026

- Won at **Canada's largest AI hackathon** by building an AI system that unifies employee tools for new hires.
- Connected **Slack** and **Github** to corporate AI using **Composio SDK** to unify business intelligence.
- Collaborated in a **4-person team** using structured Github pull requests, and deployed the MVP in **36 hours**.

**Hackathon Winner – AI Road Damage Detection** | *Python, HuggingFace, OpenCV, Supabase, Next.js*

Feb 2026

- Deployed a pretrained **YOLOv8** model from HuggingFace detecting **4 damage types** through computer vision.
- Used **Supabase** to store damage location, size, type, confidence scores, and process deduplication.
- Built a **Next.js** dashboard with **MapLibre GL** for filtering by damage type and severity across inspected roads.

**Natural Language to SQL Agent** | *Python, DuckDB, Ollama, SQL*

Mar 2026

- Built an AI tool that lets users query a database in plain English, eliminating the need to write SQL manually.
- Achieved **98% accuracy** across **172 test cases**, including trick questions and edge cases designed to break the system.
- Ran entirely on **CPU** locally through **Ollama** with no external API, keeping data private and **reducing cost to zero**.

**Custom Server Infrastructure** | *Ubuntu, Docker, Nginx, CI/CD, Github Actions*

December 2025

- Set up an **Ubuntu VPS** using **Docker** and **Nginx** to host personal and client projects concurrently.
- Hosted **5+ websites** and APIs over HTTPS, handling over **8000 unique visitors** and **300k requests monthly**.
- Built **CI/CD pipelines** with **Github Actions** to automate builds, tests, and releases across **6 Git repositories**.